Dynamic Predictive Model for Growth of *Salmonella* Enteritidis in Egg Yolk

Gompertz Model with Modified Ratkowsky Secondary Model

Member A, Member B, Member C

College of Biosystems Engineering and Food Science
Zhejiang University

April 21, 2026

Outline

1 Introduction

2 Model

3 Data & Parameters

4 Forward Problem

5 OLS Estimation

6 Residual Analysis

7 Optimal Experimental Design

8 Bootstrap

9 Summary

10 References

Introduction & Background

- *Salmonella* Enteritidis (SE) is a leading cause of foodborne illness worldwide
- Shell eggs are a major vehicle — SE penetrates shell and grows rapidly in iron-rich yolk
- USDA-FSIS requires eggs stored $\leq 7.2\,^{\circ}$C within 12 h of laying
- Temperature fluctuates during cooling, storage, and distribution
- **Objective:** Develop a dynamic Gompertz model for SE growth under sinusoidal temperature (3–43 °C, 24 h cycle)
- Reference: Gumudavelli et al. (2007), *J. Food Sci.* 72:M254–62

Model Description

Primary Model (Gompertz)

Constant T (analytical):

$$y(t) = A + C \cdot e^{-e^{(\mu e/C)(M-t)+1}}$$

Variable T (ODE):

$$\frac{dy}{dt} = -\mu(T)\, e\, \frac{y-A}{C}\, \ln\left(\frac{y-A}{C}\right)$$

$y = \log_{10} N$; A: initial; C: range; M: inflection time

Secondary Model (Ratkowsky)

$$\mu(T) = a(T - T_{\min})^2 \left(1 - e^{b(T - T_{\max})}\right)$$

a, b: regression coefficients

$T_{\min} = 6\,°\mathrm{C}$, $T_{\max} = 46.3\,°\mathrm{C}$ (fixed)

Dynamic Model

Integrate primary + secondary, solve numerically with `ode45`.

Temperature interpolated via `interp1`.

Data & Parameters

Data Summary

- Growth: 12 time points (0–17 hr), 20 observations
- 0–2 hr: single; 3–17 hr: duplicates
- Temperature: 47 points, sinusoidal ∼7–43 °C
- Interpolated via `interp1`

Param	Description	Guess	Status
A	Init. $\log_{10}$(CFU/mL)	2.60	Est.
C	Growth range	11	Est.
M	Inflection time (hr)	7.5	Est.
a	Ratkowsky coeff	3.38e-4	Est.
b	Ratkowsky coeff	0.275	Est.
$T_{\min}$	Min growth temp	6.0 °C	Fixed
$T_{\max}$	Max growth temp	46.3 °C	Fixed

Forward Problem & Scaled Sensitivity Coefficients

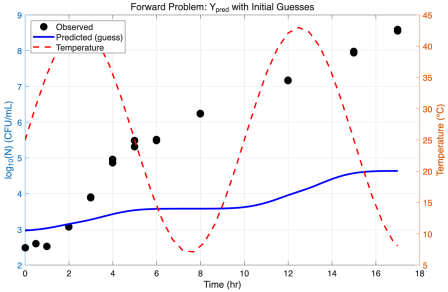

Fig. 1. Forward prediction with initial parameter guesses. Black dots: observed; blue line: predicted; red dashed: temperature.

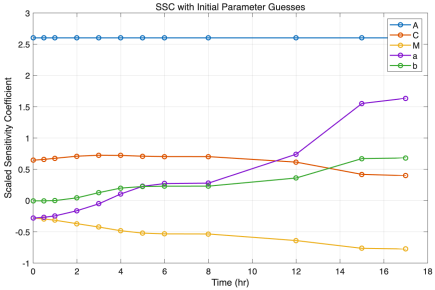

Fig. 2. Scaled sensitivity coefficients (SSC) computed with initial guesses. All 5 parameters show distinct patterns.

$T_{\min}$, $T_{\max}$ have negligible SSC $\rightarrow$ fixed at literature values.

OLS Parameter Estimates

Param	Estimate	SE	Rel.Err (%)	CI_{lo}	CI_{hi}
A	...	...	...	...	...
C	...	...	...	...	...
M	...	...	...	...	...
a	...	...	...	...	...
b	...	...	...	...	...

	A	C	M	a	b
A	1	...	...	...	...
C		1	...	...	...
M			1	...	...
a				1	...
b					1

Metric	Value
RMSE	...
Pseudo-R^2	...
MSE	...
N	20

Fitted Curve with Confidence & Prediction Bands

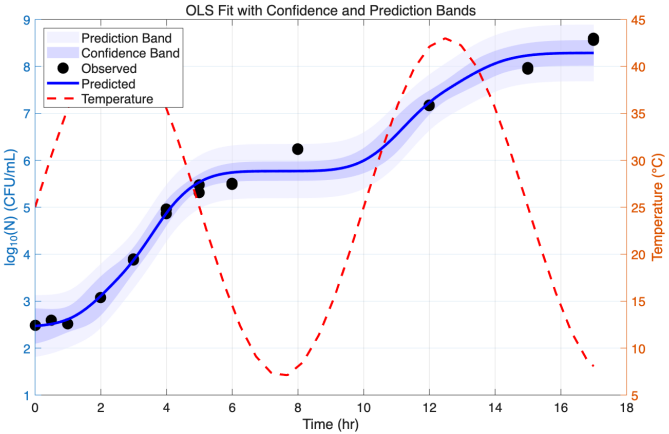

Fig. 3. OLS fitted curve (blue) with 95% asymptotic confidence band (dark shading) and prediction band (light shading). Black dots: observed data; red dashed: sinusoidal temperature profile.

Residual Analysis

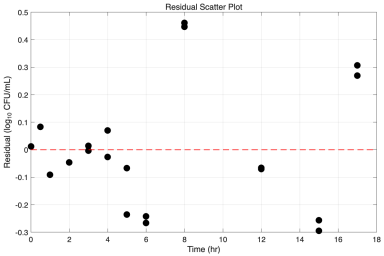

Fig. 4. Residuals vs. time. Random scatter around zero indicates no systematic bias.

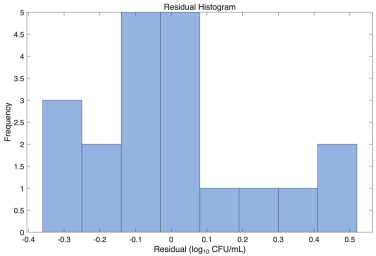

Fig. 5. Residual histogram. Approximately bell-shaped, supporting normality assumption.

#	Assumption	Result
1	Model is correct	✓ (visual)
2	Errors are random	DW = ...
3	Constant variance	✓ (visual)
4	Errors uncorrelated	See DW
5	Normal distribution	SW p = ...

Final Scaled Sensitivity Coefficients

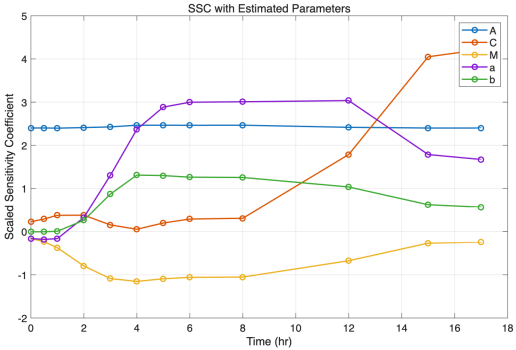

Fig. 6. SSC recomputed with estimated parameters. Patterns are consistent with initial SSC, confirming identifiability.

Param	max|SSC|
A	...
C	...
M	...
a	...
b	...

All parameters remain identifiable at the OLS solution.

Optimal Experimental Design

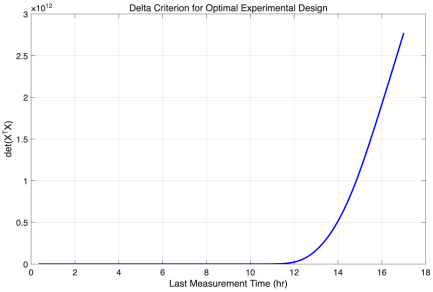

Fig. 7. Delta criterion $\det(X^T X)$ vs. last measurement time. Increasing trend indicates more data improves estimation.

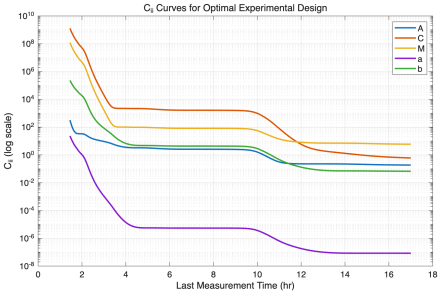

Fig. 8. C_{ii} curves (log scale). Decreasing values indicate improved parameter precision with more measurements.

Key measurement window: exponential growth phase (∼2–8 hr).

Bootstrap Analysis ($N = 1000$)

Param	Bootstrap CI Lower	Bootstrap CI Upper	Asymptotic CI Lower	Asymptotic CI Upper
A	...	...	...	...
C	...	...	...	...
M	...	...	...	...
a	...	...	...	...
b	...	...	...	...

Band	Avg Width
Asymptotic CB	...
Bootstrap CB	...
Asymptotic PB	...
Bootstrap PB	...

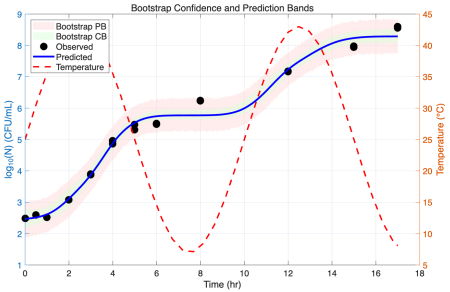

Fig. 9. Bootstrap confidence band (green) and prediction band (pink) from 1000 residual-resampling iterations.

Summary & Conclusions

- Dynamic Gompertz model successfully developed for SE growth in egg yolk
- 5 parameters estimated, 2 fixed; RMSE $< 0.3\ \log_{10}$ CFU/mL
- All 5 standard statistical assumptions evaluated
- Optimal experimental design: key window at 2–8 hr
- Bootstrap CIs consistent with asymptotic CIs
- **Practical significance:** Model predicts SE risk during egg cooling, storage, and distribution — supports USDA-FSIS 7.2 °C recommendation

References

- Gumudavelli V, Subbiah J, Thippareddi H, Velugoti PR, Froning G. 2007. Dynamic predictive model for growth of *Salmonella* Enteritidis in egg yolk. *J. Food Sci.* 72:M254–62.
- Baranyi J, Roberts TA. 1994. A dynamic approach to predicting bacterial growth in food. *Int J Food Microbiol* 23:277–94.
- Zwietering MH et al. 1991. Modeling of the bacterial growth as a function of temperature. *Appl Environ Microbiol* 57:1094–101.
- Beck JV, Arnold KJ. 1977. *Parameter Estimation in Engineering and Science*. Wiley.

Thank You!

Questions?